

RITIKA KAUSHIK

+91-7078988998 ✉ rkaushik_be23@thapar.edu  LinkedIn  Github  LeetCode

EDUCATION

Thapar Institute Of Engineering & Technology, Patiala	Aug 2023 - Present
B.Tech in Computer Engineering	CGPA: 9.64
St. Mary's Academy, Meerut	Apr 2022 - Mar 2023
Senior Secondary School certificate	Percentage: 97%
St. Mary's Academy, Meerut	Apr 2020 - Mar 2021
Senior Secondary School certificate	Percentage: 97%

PROJECTS

UniVerse - CampusGPT Assistant for TIET Students



- Built a hybrid AI-powered campus assistant for Thapar University, enabling real-time student queries across academics, administration, dining, healthcare, and campus navigation using LLM-based intent detection integrated with MySQL and JSON-backend services.
- Designed a graph based campus navigation system using weighted nodes and edges, implementing Dijkstra's algorithm with Leaflet+OpenStreetMap visualization and OSRM-based routing for distance-optimized walking paths.
- **Tech Stack** - MySQL, Google Gemini API, Vercel, Render, Python

Bangalore Traffic Prediction



- Built a Bangalore traffic congestion prediction system using historical traffic, weather, environmental impact, and temporal features, deployed via a FastAPI backend to deliver real-time congestion levels, categories, and analytical insights.
- Engineered spatial-temporal features and trained a stacked Random Forest-XGBoost regression model, achieving RMSE = 2.031 and $R^2 = 0.995$, outperforming baseline models and prior studies.
- **Tech Stack** - Python, FastAPI, Google Gemini API, HTML, CSS

CaloriesTracker



- Developed an AI-powered calorie tracking web app that analyzes uploaded food images to estimate calorie intake, nutritional composition, and associated health insights using Google Gemini Pro Vision.
- Designed a multimodal pipeline integrating computer vision and LLM-based reasoning to recognize food items and generate structured dietary summaries, including portion recommendations and potential health considerations.
- **Tech Stack** - Python, Generative AI, Streamlit

TECHNICAL SKILLS

Languages: Python, C, C++, Java

Libraries & Frameworks: LangChain, Hugging Face Transformers, NumPy, Pandas, Scikit-learn, FastAPI

Artificial Intelligence/Machine Learning: Generative AI, Large Language Models, Retrieval Augmented Generation

Tools & Technologies: MySQL, Oracle, Chroma, HTML/CSS, Ollama, Vercel, Linux, VS Code, Render, GitHub

Core CS Concepts: Object-Oriented Programming (OOP), Operating Systems (OS), DBMS, Data Structures and Algorithms, Computer Networks, Artificial Intelligence

ACHIEVEMENTS

Awarded 100% tuition scholarship for consistent academic excellence.

Gold Medalist in National Mathematics Olympiad, competing among thousands of participants worldwide.

Best All-Rounder Award, Class XII

CERTIFICATION

Software Engineer Intern (HackerRank)

Complete Data Science, Machine Learning, DL, NLP Bootcamp (IN-PROGRESS)

Professional Kathak Dance Certification (8 years off formal training)